Create Database and Import Data

```
use salesData

db.createCollection("sales")

db.sales.insertMany([
{
"saleDate": ISODate("2023-01-15T00:00:00Z"),
"customer": { "customerId": "C1001", "name": "Alice Smith" },
"product": { "productId": "P201", "name": "Laptop", "category": "Electronics" },
"amount": 1200,
"quantity": 1
},
{
"saleDate": ISODate("2023-02-20T00:00:00Z"),
"customer": { "customerId": "C1002", "name": "Bob Johnson" },
"product": { "productId": "P202", "name": "Tablet", "category": "Electronics" },
"amount": 300,
"quantity": 2
},
{
"saleDate": ISODate("2023-03-10T00:00:00Z"),
"customer": { "customerId": "C1001", "name": "Alice Smith" },
"product": { "productId": "P301", "name": "T-shirt", "category": "Clothing" },
"amount": 25,
"quantity": 5
},
{
"saleDate": ISODate("2023-04-05T00:00:00Z"),
"customer": { "customerId": "C1003", "name": "Charlie Brown" },
"product": { "productId": "P201", "name": "Laptop", "category": "Electronics" },
"amount": 1100,
"quantity": 1
},
{
"saleDate": ISODate("2023-05-12T00:00:00Z"),
"customer": { "customerId": "C1002", "name": "Bob Johnson" },
"product": { "productId": "P302", "name": "Jeans", "category": "Clothing" },
"amount": 60,
"quantity": 1
},
{
"saleDate": ISODate("2023-06-22T00:00:00Z"),
```

"customer": { "customerId": "C1004", "name": "David Lee" },
"product": { "productId": "P401", "name": "Coffee Maker", "category": "Appliances" },
"amount": 80,
"quantity": 1
},
{
"saleDate": ISODate("2023-07-01T00:00:00Z"),
"customer": { "customerId": "C1001", "name": "Alice Smith" },
"product": { "productId": "P202", "name": "Tablet", "category": "Electronics" },
"amount": 320,
"quantity": 2
},
{
"saleDate": ISODate("2023-08-18T00:00:00Z"),
"customer": { "customerId": "C1003", "name": "Charlie Brown" },
"product": { "productId": "P301", "name": "T-shirt", "category": "Clothing" },
"amount": 30,
"quantity": 6
},
{
"saleDate": ISODate("2023-09-05T00:00:00Z"),
"customer": { "customerId": "C1005", "name": "Eve Wilson" },
"product": { "productId": "P201", "name": "Laptop", "category": "Electronics" },
"amount": 1250,
"quantity": 1
},
{
"saleDate": ISODate("2023-10-11T00:00:00Z"),
"customer": { "customerId": "C1004", "name": "David Lee" },
"product": { "productId": "P302", "name": "Jeans", "category": "Clothing" },
"amount": 70,
"quantity": 1
}
])

```
db.sales.find().pretty()
```

Step 1: Calculate Total Sales per Customer

Description

This pipeline calculates the total amount spent by each customer, grouping all transactions by their customerId. It helps identify which customers contribute most to overall revenue.

Command: db.sales.aggregate([{ $group: { _id: "$customer.customerId", totalSales: { $sum: "$amount" }, customerName: { $first: "$customer.name" } } }, { $project: { _id: 0, customerId: "$_id", customerName: 1, totalSales: 1 } }, { $sort: { totalSales: -1 } }])

```
salesData> db.sales.aggregate([ { $group: { _id: "$customer.customerId", totalSales: { $sum: "$amount" }, customerName: { $first: "
$customer.name" } } }, { $project: { _id: 0, customerId: "$_id", customerName: 1, totalSales: 1 } }, { $sort: { totalSales: -1 } }
])
[
  { totalSales: 1545, customerName: 'Alice Smith', customerId: 'C1001' },
  { totalSales: 1250, customerName: 'Eve Wilson', customerId: 'C1005' },
  {
    totalSales: 1130,
    customerName: 'Charlie Brown',
    customerId: 'C1003'
  },
  { totalSales: 360, customerName: 'Bob Johnson', customerId: 'C1002' },
  { totalSales: 150, customerName: 'David Lee', customerId: 'C1004' },
  { totalSales: 0, customerName: null, customerId: null }
]
```

Insight

Alice Smith emerges as the highest spender, indicating strong loyalty and potential for targeted promotions or rewards.

Step 2: Find the Most Popular Product Category

Description

This aggregation identifies the product category with the highest number of items sold, allowing a clear view of consumer preferences in the store's inventory.

Command: db.sales.aggregate([{ $group: { _id: "$product.category", totalQuantity: { $sum: "$quantity" } } }, { $sort: { totalQuantity: -1 } }, { $limit: 1 }, { $project: { _id: 0, category: "$_id", totalQuantity: 1 } }])

```
salesData> db.sales.aggregate([ { $group: { _id: "$product.category", totalQuantity: { $sum: "$quantity" } } }, { $sort: { totalQua
totalQuantity: -1 } }, { $limit: 1 }, { $project: { _id: 0, category: "$_id", totalQuantity: 1 } } ])
[ { totalQuantity: 13, category: 'Clothing' } ]
```

Insight

The Clothing category leads in quantity sold, suggesting that apparel products have the strongest market demand among all categories.

Step 3: Analyze Sales by Month

Description

This pipeline sums sales for each month to reveal seasonal trends and understand fluctuations in revenue over the year.

Command: db.sales.aggregate([{ $group: { _id: { $month: "$saleDate" }, totalSales: { $sum: "$amount" } } }, { $project: { _id: 0, month: "$_id", totalSales: 1 } }, { $sort: { month: 1 } }])

```
salesData> db.sales.aggregate([ { $group: { _id: { $month: "$saleDate" }, totalSales: { $sum: "$amount" } } }, { $project: { _id: 0
, month: "$_id", totalSales: 1 } }, { $sort: { month: 1 } } ])
[
  { totalSales: 0, month: null },
  { totalSales: 1200, month: 1 },
  { totalSales: 300, month: 2 },
  { totalSales: 25, month: 3 },
  { totalSales: 1100, month: 4 },
  { totalSales: 60, month: 5 },
  { totalSales: 80, month: 6 },
  { totalSales: 320, month: 7 },
  { totalSales: 30, month: 8 },
  { totalSales: 1250, month: 9 },
  { totalSales: 70, month: 10 }
]
```

Insight

Sales peak in September, which may indicate higher consumer activity or successful marketing campaigns during that period.

Step 4: Find Customers Who Purchased Laptops

Description

This pipeline filters the dataset to identify customers who bought laptops. It helps in segmenting customers by product preference.

Command: db.sales.aggregate([{ $match: { "product.name": "Laptop" } }, { $group: { _id: "$customer.customerId", customerName: { $first: "$customer.name" } } }, { $project: { _id: 0, customerId: "$_id", customerName: 1 } }])

```
salesData> db.sales.aggregate([ { $match: { "product.name": "Laptop" } }, { $group: { _id: "$customer.customerId", customerName: {
$first: "$customer.name" } } }, { $project: { _id: 0, customerId: "$_id", customerName: 1 } } ])
[
  { customerName: 'Eve Wilson', customerId: 'C1005' },
  { customerName: 'Alice Smith', customerId: 'C1001' },
  { customerName: 'Charlie Brown', customerId: 'C1003' }
]
```

Insight

Three customers—Alice Smith, Charlie Brown, and Eve Wilson—purchased laptops, showing a smaller but high-value segment interested in electronics.

Step 5: Calculate Average Purchase Amount

Description

This aggregation calculates the mean transaction value across all sales, providing a benchmark for customer spending habits.

Command: db.sales.aggregate([{ $group: { _id: null, averageAmount: { $avg: "$amount" } } }, { $project: { _id: 0, averageAmount: 1 } }])

```
salesData> db.sales.aggregate([ { $group: { _id: null, averageAmount: { $avg: "$amount" } } }, { $project: { _id: 0, averageAmount:
 1 } } ])
[ { averageAmount: 443.5 } ]
```

Insight

The average purchase amount is 473.5, which can guide pricing strategies and promotional offers to optimize revenue.